Lösung: Ableitungen

- $f(x) = 4x^3$

  $f'(x) = 4 \cdot 3x^2 = 12x^2$

- $f(x) = 4x^3 + x^2 + x + 1$

  $f'(x) = 4 \cdot 3x^2 + 2x + 1 + 0 = 12x^2 + 2x + 1$

- $f(x) = \sqrt{x} = x^{1/2}$

  $f'(x) = 1/2\, x^{-1/2} = \frac{1}{2x^{1/2}} = \frac{1}{2\sqrt{x}}$

- $f(x) = \sqrt[4]{x^3} = (x^3)^{1/4} = x^{3/4}$

  $f'(x) = \frac{3}{4} x^{3/4-1} = \frac{3}{4} x^{-1/4} = \frac{3}{4} \frac{1}{x^{1/4}} = \frac{3}{4\sqrt[4]{x}}$

- $f(x) = t^2 e^{3t}$

  $f'(x) = t^2 (e^{3t})' + (t^2)' \cdot e^{3t} = t^2 \cdot e^{3t} \cdot 3 + 2t\, e^{3t}$

  $= e^{3t} (3t^2 + 2t)$

- $f(x) = (x^n + 1) \cdot \sqrt{x^2 - 1}$

  $f'(x) = (x^n + 1)' \cdot \sqrt{x^2-1} + (x^n + 1) \left((x^2-1)^{1/2}\right)'$

  $= n \cdot x^{n-1} \cdot \sqrt{x^2-1} + (x^n + 1) \cdot \frac{1}{2} (x^2-1)^{-1/2} \cdot 2x$

  $= n \cdot x^{n-1} \cdot \sqrt{x^2-1} + (x^n + 1) \frac{x}{\sqrt{x^2-1}}$

  $= \frac{n x^{n-1} (x^2-1) + (x^n + 1) \cdot x}{\sqrt{x^2-1}}$

  $= \frac{n x^{n+1} - n x^{n-1} + x^{n+1} + x}{\sqrt{x^2-1}}$

  $= \frac{(n+1) x^{n+1} - n x^{n-1} + x}{\sqrt{x^2-1}}$

- $f(x) = \frac{10x}{x^2+1}$

$$f'(x) = \frac{(x^2+1)\cdot(10x)' - (x^2+1)'\cdot 10x}{(x^2+1)^2} = \frac{(x^2+1)\cdot 10 - 2x\cdot 10x}{(x^2+1)^2}$$

$$= \frac{10x^2+10-20x^2}{(x^2+1)^2} = \frac{10-10x^2}{(x^2+1)^2} = 10\,\frac{1-x^2}{(1+x^2)^2}$$

- $f(x) = \frac{1+x}{1+x^2}$

$$f'(x) = \frac{(1+x^2)\cdot(1+x)' - (1+x^2)'\cdot(1+x)}{(1+x^2)^2} = \frac{(1+x^2)\cdot 1 - 2x\cdot(1+x)}{(1+x^2)^2}$$

$$= \frac{1+x^2-2x-2x^2}{(1+x^2)^2} = \frac{1-x^2-2x}{(1+x^2)^2}$$

- $f(x) = 5\,(x^3-x^2+1)^5$

$$f'(x) = 5\cdot 5\,(x^3-x^2+1)^4\cdot(3x^2-2x+0)$$

$$= 25\,(x^3-x^2+1)^4\cdot(3x^2-2x)$$

- $f(x) = \left(\frac{1+x}{x}\right)^n$

$$f'(x) = n\left(\frac{1+x}{x}\right)^{n-1}\cdot\left(\frac{1+x}{x}\right)'$$

$$= n\left(\frac{1+x}{x}\right)^{n-1}\cdot\frac{x(1+x)' - (1+x)(x)'}{x^2}$$

$$= n\left(\frac{1+x}{x}\right)^{n-1}\cdot\frac{x\cdot 1 - (1+x)\cdot 1}{x^2}$$

$$= n\left(\frac{1+x}{x}\right)^{n-1}\cdot\frac{\cancel{x}-1-\cancel{x}}{x^2} = -\frac{n}{x^2}\cdot\frac{(1+x)^{n-1}}{x^{n-1}} = -n\cdot\frac{(1+x)^{n-1}}{x^{n+1}}$$

- $f(x) = 2x \cdot \sqrt{x^2-1}$

$$f'(x) = (2x)' \cdot \sqrt{x^2-1} + 2x \cdot \underbrace{\left(\sqrt{x^2-1}\right)'}_{(x^2-1)^{1/2}}$$

$$= 2 \cdot \sqrt{x^2-1} + 2x \cdot \tfrac{1}{2} \cdot (x^2-1)^{-1/2} \cdot 2x$$

$$= 2 \cdot \sqrt{x^2-1} + \frac{2x^2}{\sqrt{x^2-1}} = \frac{2}{\sqrt{x^2-1}} \left(x^2-1+x^2\right)$$

$$= \frac{2\,(2x^2-1)}{\sqrt{x^2-1}}$$

- $u(t) = u_0 \cdot \sin(\omega t + \varphi)$

$$u'(t) = u_0 \cdot \cos(\omega t+\varphi) \cdot (\omega t+\varphi)'$$

$$= u_0 \cdot \cos(\omega t+\varphi) \cdot \omega = u_0 \cdot \omega \cdot \cos(\omega t+\varphi)$$

- $f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)}$

$$f'(x) = \frac{\cos(x)\,(\sin(x))' - (\cos(x))' \cdot \sin(x)}{\cos^2(x)}$$

$$= \frac{\cos(x)\cos(x) - (-\sin(x)) \cdot \sin(x)}{\cos^2(x)}$$

$$= \frac{\cos^2(x) + \sin^2(x)}{\cos^2(x)} = \underline{\underline{\frac{1}{\cos^2(x)}}}$$

(oder:

$$= \frac{\cos^2(x)}{\cos^2(x)} + \frac{\sin^2(x)}{\cos^2(x)} = 1 + \left(\frac{\sin(x)}{\cos(x)}\right)^2$$

$$= \underline{\underline{1 + \tan^2(x)}}$$

Daraus folgt auch die trigonometrische Relation

$$\underline{\underline{\frac{1}{\cos^2(x)} = 1 + \tan^2(x)}}$$

- $f(x) = a^x = \left(e^{\ln(a)}\right)^x = e^{x \cdot \ln(a)}$

$f'(x) = e^{x \cdot \ln(a)} \cdot (x \cdot \ln(a))' = e^{x \cdot \ln(a)} \cdot \ln(a)$

$= a^x \cdot \ln(a)$

- $f(x) = \sqrt{a^2 - x^2} = (a^2 - x^2)^{1/2}$

$f'(x) = \frac{1}{2}(a^2 - x^2)^{-1/2} \cdot (a^2 - x^2)' = \frac{1}{2} \cdot (a^2 - x^2)^{-1/2} \cdot (-2x)$

$= -\frac{x}{\sqrt{a^2 - x^2}}$

- $f(x) = \sin(2x)$

$f'(x) = \cos(2x) \cdot (2x)' = \cos(2x) \cdot 2$

- $f(x) = \frac{a}{x}\sqrt{a^2 + x^2} = a x^{-1}(a^2 + x^2)^{1/2}$

$f'(x) = (ax^{-1})' \cdot (a^2 + x^2)^{1/2} + (ax^{-1}) \cdot \left((a^2 + x^2)^{1/2}\right)'$

$= -ax^{-2} \cdot (a^2 + x^2)^{1/2} + ax^{-1} \cdot \frac{1}{2}(a^2 + x^2)^{-1/2} \cdot 2x$

$= -a\frac{\sqrt{a^2 + x^2}}{x^2} + a\frac{1}{\sqrt{a^2 + x^2}} = \frac{a}{\sqrt{a^2 + x^2}}\left(1 - \frac{a^2 + x^2}{x^2}\right)$

$= \frac{a}{\sqrt{a^2 + x^2}}\left(\frac{x^2 - a^2 - x^2}{x^2}\right) = \frac{-a^3}{x^2\sqrt{a^2 + x^2}}$

- $f(x) = \frac{1}{\sin(x)} = (\sin(x))^{-1}$

$f'(x) = (-1) \cdot (\sin(x))^{-2} \cdot \underbrace{(\sin(x))'}_{= \cos(x)}$

$= \frac{-\cos(x)}{\sin^2(x)} = \frac{-1}{\sin(x) \cdot \tan(x)}$

- $f(x) = \cos^3(x) = (\cos(x))^3$

  $f'(x) = 3\,(\cos(x))^2 \cdot (\cos(x))' = -3 \cdot \cos^2(x) \cdot \sin(x)$

- $f(x) = \sin(x^3)$

  $f'(x) = \cos(x^3) \cdot (x^3)' = \cos(x^3) \cdot 3x^2$

- $f(x) = \tan^3(x) - 3\tan(x)$

  $f'(x) = 3\tan^2(x) \cdot (\tan(x))' - 3 \cdot (\tan(x))'$

  $= 3 \cdot \tan^2(x)\,(1 + \tan^2(x)) - 3\,(1 + \tan^2(x))$

  $= 3\tan^2(x) + 3 \cdot \tan^4(x) - 3 - 3\tan^2(x)$

  $= 3 \cdot (\tan^4(x) - 1)$

- $f(x) = \tan(1/x)$

  $f'(x) = \frac{1}{\cos^2(1/x)} \cdot (1/x)' = \frac{-1}{x^2 \cdot \cos^2(1/x)} = \underline{\underline{-(x \cdot \cos(1/x))^{-2}}}$

- $f(x) = \frac{x^2 - a^2}{\sin(x)}$

  $f'(x) = \frac{\sin(x)\,(x^2-a^2)' - (\sin(x))' \cdot (x^2-a^2)}{\sin^2(x)} = \frac{\sin(x)\,2x - \cos(x)\,(x^2-a^2)}{\sin^2(x)}$

- $f(x) = \tan(x^2) \cdot x^2$

  $f'(x) = (\tan(x^2))' \cdot x^2 + \tan(x^2) \cdot (x^2)'$

  $= (1 + \tan^2(x^2)) \cdot 2x \cdot x^2 + \tan(x^2) \cdot 2x$

  $= (1 + \tan^2(x^2))\,2x^3 + \tan(x^2) \cdot 2x$

- $f(x) = 3x + \sin(\cos(x))$

  $f'(x) = 3 + \cos(\cos(x)) \cdot (\cos(x))'$

  $= 3 + \cos(\cos(x)) \cdot (-\sin(x))$

  $= 3 - \cos(\cos(x)) \cdot \sin(x)$

- $f(x) = \sqrt[5]{x^4 - x^2} = (x^4 - x^2)^{1/5}$

  $f'(x) = 1/5\,(x^4 - x^2)^{-4/5} \cdot (4x^3 - 2x) = \dfrac{2x(2x^2-1)}{5\sqrt[5]{(x^4-x^2)^4}}$

- $f(x) = \sqrt[5]{(x-6)^4} = (x-6)^{4/5}$

  $f'(x) = 4/5\,(x-6)^{-1/5} \cdot 1 = \dfrac{4}{5 \cdot \sqrt[5]{x-6}}$

- $f(x) = e^{-x} \cdot x^2$

  $f'(x) = (e^{-x})' \cdot x^2 + e^{-x} \cdot (x^2)' = -e^{-x} x^2 + e^{-x} \cdot 2x$

  $= e^{-x}(2x - x^2) = e^{-x} \cdot x \cdot (2-x)$